FIBRE PRODUCTS OVER A CATEGORY FROM STRONGLY CARTESIAN LIFTS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 6546-6575, following the statement at 6530-6544.

Editorial note. The proof is complete with no omission markers; the middle equalities of the chain are set-level pullback reassociations (display-data level).

Allowed standard prerequisites (editorial).

- Definition of strongly cartesian morphism. definition-cartesian-over-C and the discussion following it, invoked at proof lines 6572-6574. Supplies Mor_S(t,x) = Mor_S(t,y) x_{Mor_C} Mor_C for the strongly cartesian morphisms x -> y and a: w -> z, justifying the first and last equalities of the chain.
- Set level pullback algebra. the middle equalities of the chain at proof lines 6550-6571. Reassociation of iterated pullbacks of sets and insertion of the identity factor Mor_S(t,y).

Lemma 0.1. *Let $\mathcal{C}$ be a category. Let $p : \mathcal{S} \to \mathcal{C}$ be a category over $\mathcal{C}$. Let $x \to y$ and $z \to y$ be morphisms of $\mathcal{S}$. Assume*

(1) *$x \to y$ is strongly cartesian,*
(2) *$p(x) \times_{p(y)} p(z)$ exists, and*
(3) *there exists a strongly cartesian morphism $a : w \to z$ in $\mathcal{S}$ with $p(w) = p(x) \times_{p(y)} p(z)$ and $p(a) = pr_2 : p(x) \times_{p(y)} p(z) \to p(z)$.*

Then the fibre product $x \times_y z$ exists and is isomorphic to w.

Proof. Since $x \to y$ is strongly cartesian there exists a unique morphism $b : w \to x$ such that $p(b) = \mathrm{pr}_1$. To see that w is the fibre product we compute

$$\begin{aligned}
&\mathrm{Mor}_{\mathcal{S}}(t, w) \\
&= \mathrm{Mor}_{\mathcal{S}}(t, z) \times_{\mathrm{Mor}_{\mathcal{C}}(p(t), p(z))} \mathrm{Mor}_{\mathcal{C}}(p(t), p(w)) \\
&= \mathrm{Mor}_{\mathcal{S}}(t, z) \times_{\mathrm{Mor}_{\mathcal{C}}(p(t), p(z))} (\mathrm{Mor}_{\mathcal{C}}(p(t), p(x)) \times_{\mathrm{Mor}_{\mathcal{C}}(p(t), p(y))} \mathrm{Mor}_{\mathcal{C}}(p(t), p(z))) \\
&= \mathrm{Mor}_{\mathcal{S}}(t, z) \times_{\mathrm{Mor}_{\mathcal{C}}(p(t), p(y))} \mathrm{Mor}_{\mathcal{C}}(p(t), p(x)) \\
&= \mathrm{Mor}_{\mathcal{S}}(t, z) \times_{\mathrm{Mor}_{\mathcal{S}}(t, y)} \mathrm{Mor}_{\mathcal{S}}(t, y) \times_{\mathrm{Mor}_{\mathcal{C}}(p(t), p(y))} \mathrm{Mor}_{\mathcal{C}}(p(t), p(x)) \\
&= \mathrm{Mor}_{\mathcal{S}}(t, z) \times_{\mathrm{Mor}_{\mathcal{S}}(t, y)} \mathrm{Mor}_{\mathcal{S}}(t, x)
\end{aligned}$$

as desired. The first equality holds because $a : w \to z$ is strongly cartesian and the last equality holds because $x \to y$ is strongly cartesian. □

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